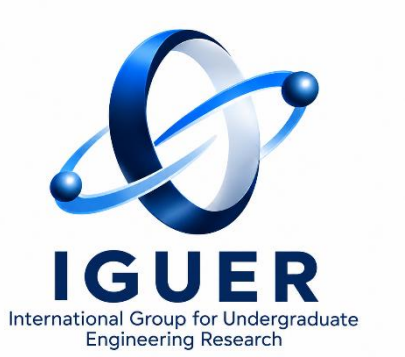


Computational State Point: Conceptual Development, Methodological Limitations, and Suspension of Research

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Abstract

The Computational State Point (CSP) framework was proposed as an alternative approach to computational fluid dynamics, aiming to represent fluid behavior through discrete computational points and their geometric interactions. Over approximately seven months, the research explored network-based relationships, virtual volume reconstruction, and energy-based equations of motion. However, the framework remained mathematically incomplete, and its conservation properties, numerical stability, and shock-capturing performance were not sufficiently validated. A subsequent reassessment of established numerical methods also revealed insufficient evidence for the anticipated advantages of CSP. Consequently, the research was suspended in October 2026. This report documents the conceptual development, unresolved methodological limitations, and rationale for suspending the project.

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Abstract

The Computational State Point (CSP) framework was proposed as an alternative approach to computational fluid dynamics, aiming to represent fluid behavior through discrete computational points and their geometric interactions. Over approximately seven months, the research explored network-based relationships, virtual volume reconstruction, and energy-based equations of motion. However, the framework remained mathematically incomplete, and its conservation properties, numerical stability, and shock-capturing performance were not sufficiently validated. A subsequent reassessment of established numerical methods also revealed insufficient evidence for the anticipated advantages of CSP. Consequently, the research was suspended in October 2026. This report documents the conceptual development, unresolved methodological limitations, and rationale for suspending the project.

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1. Introduction

Computational fluid dynamics (CFD) is widely employed to investigate compressible fluid motion, including the propagation of shock waves, expansion waves, and contact discontinuities. Numerical methods based on the Euler or Navier–Stokes equations provide established approaches to these problems, while particle-based and mesh-free methods offer alternative representations of fluid motion [1, 2, 3].

Nevertheless, numerical difficulties may arise when resolving discontinuities, maintaining stability, or balancing numerical diffusion against spatial resolution. The extent and significance of these limitations depend on the governing equations, discretization scheme, numerical flux, stabilization treatment, and problem configuration.

The Computational State Point (CSP) framework was initiated to explore a different computational representation of fluid motion. Rather than defining the fluid state exclusively through a conventional spatial mesh, the research considered a collection of discrete points whose geometric relationships could be used to reconstruct relevant physical quantities and determine their evolution.

An important motivation was the investigation of compressible flows in wave disk engines (WDEs), where the propagation and interaction of pressure waves are central to engine operation. The initial hypothesis was that a discrete-point formulation might offer advantages

in representing material motion and discontinuities.

However, the hypothesis required both theoretical justification and numerical demonstration. The existence of a different computational representation does not, by itself, establish greater accuracy, stability, or efficiency.

The present report provides an archival account of CSP development and explains why further investigation was suspended before these claims could be validated.

2. Conceptual Development of the CSP Framework

2.1. Fundamental Representation

The initial framework was developed around geometric space, time, and discrete computational entities.

Geometric space was represented by a domain Ω in which positions and geometric relationships could be defined. Time was introduced as a continuous parameter describing the evolution of the system.

Each computational point was associated with a position, an invariant mass, and physical state information. Unlike a conventional finite-volume cell, a point was not assumed to possess a fixed intrinsic volume.

In earlier formulations, these entities were described as material elements. Subsequent conceptual revisions placed greater emphasis on their interpretation as discrete computational points rather than independently defined physical volumes.

The absence of an intrinsic volume created an immediate requirement: quantities such as density and pressure could not be evaluated without an additional reconstruction mechanism. This requirement motivated the introduction of the Network and Virtual Volume concepts.

2.2. Network-Based Geometric Relationships

The Network was introduced to describe geometric relationships between computational points. Several candidate formulations were considered, including inverse-distance relationships and kernel-based weighting schemes. Their purpose was to determine how nearby points contributed to quantities required for fluid-state reconstruction.

For a one-dimensional configuration, an exploratory inverse-distance formulation was considered:

$$N_i = \left(\frac{1}{\Delta x_{i-1}} + \frac{1}{\Delta x_{i+1}}\right)^{-1}$$

where the distances describe the separations between point (i) and its neighboring points. This expression was investigated as a geometric measure associated with local point distribution. It was not established as a universally applicable definition of the Network.

A central unresolved issue was the selection of a network formulation that could simultaneously support physically meaningful state reconstruction, conservation properties, and stable numerical evolution.

The relationship between network selection and the resulting computational equations therefore remained an essential part of the proposed framework.

2.3. Virtual Volume

The Virtual Volume (VV) was introduced as a computational quantity for reconstructing volume-dependent physical variables.

Although individual CSPs were not assigned intrinsic physical volumes, a virtual volume could be associated with each point through the Network.

One exploratory global-volume allocation was expressed as

$$V_i = V_{total} \frac{N_i}{\sum_{k=1}^{n} N_k}$$

where V_{total} represents the prescribed total volume of the system.

This formulation guarantees that the sum of allocated virtual volumes equals the prescribed total volume, provided the denominator is nonzero. However, this algebraic property does not by itself establish physical consistency or numerical conservation under arbitrary motion and boundary conditions.

The point density could subsequently be defined as

$$\rho_i = \frac{m_i}{V_i}$$

where m_i is the mass assigned to computational point i.

The resulting density was intended to provide a connection between discrete geometry and thermodynamic state variables. However, the validity of this reconstruction under nonuniform point distributions, large deformations, discontinuities, and open-boundary conditions remained unresolved.

2.4. Thermodynamic State and Equations of Motion

The framework explored the use of density-dependent thermodynamic energy to derive the motion of computational points.

A proposed development sequence was:

1. Define the network relationships between computational points.
2. Determine virtual volumes from the network.
3. Reconstruct density and other thermodynamic quantities.
4. Construct the system's kinetic and internal energies.
5. Define a Lagrangian from the energy expressions.
6. Apply the Euler–Lagrange equations to derive equations of motion.

This approach was motivated by the possibility of obtaining equations consistent with a discrete mechanical energy formulation. Exploratory mathematical derivations were developed under restricted conditions. These investigations suggested potential relationships between network-derived volume changes and continuum-mechanical quantities. However, the derivations did not establish a complete equivalence between CSP and the compressible Euler or Navier–Stokes equations for general flows. In particular, constructing equations from a Lagrangian was not sufficient to establish numerical stability, entropy admissibility, or accurate representation of discontinuous solutions.

2.5. Alternative Computational Representations

As development progressed, alternative representations were examined to address limitations associated with network-derived volume reconstruction. These included geometric approaches involving virtual boundaries or boundary planes between computational points, as well as alternative virtual-volume definitions.

The purpose of these explorations was to investigate whether local geometric interactions could provide a more consistent representation of compression, expansion, and interaction forces. However, these alternatives introduced additional questions concerning the physical interpretation of virtual boundaries, effective inertia, force transmission, and energy conservation. No alternative formulation reached a sufficiently complete and validated state to replace the original framework. Accordingly, these proposals are retained as exploratory concepts rather than established components of a working numerical method.

3. Mathematical and Computational Assessment

3.1. Conservation Properties

Mass conservation was incorporated conceptually by assigning invariant mass to individual computational points.

$$M = \sum_{i=1}^{n} m_i$$

$$\frac{dM}{dt} = 0$$

However, invariant point mass alone does not guarantee that the complete numerical method conserves momentum and total energy. Conservation of momentum requires an appropriate treatment of internal forces, while conservation of total energy depends on the consistency of the equations of motion, thermodynamic formulation, and numerical time integration. Pairwise and symmetric interactions were therefore investigated as possible mechanisms for maintaining momentum conservation. Energy-based derivations were also explored through the Euler–Lagrange formalism.

Nevertheless, a general conservation proof covering arbitrary network configurations, boundary conditions, and numerical implementations was not completed. The distinction between conservation properties of an idealized mathematical formulation and those of a fully discrete numerical algorithm remained unresolved.

Exploratory mathematical verification was also undertaken using Lean, with implementation assistance from OpenAI Codex. The formalization covered inverse-distance and kernel-based network formulations, including conservation-related identities, pressure-force relationships, and conditional continuum-consistency statements. Counterexamples and restrictive assumptions were also examined. These formal developments are limited to the explicitly encoded mathematical statements and do not establish the physical validity or numerical reliability of the complete CSP framework (see Supplementary Material S3).

3.2. Continuum Consistency

One of the central theoretical requirements was determining whether CSP could reproduce the appropriate continuum equations under refinement. Preliminary derivations examined the relationship between point distribution, network evolution, virtual volume, density, and pressure forces.

Under restrictive assumptions, including one-dimensional configurations and simplified boundary treatment, certain expressions exhibited relationships with continuum compression and pressure-gradient terms. These observations indicated possible mathematical consistency in limited settings.

However, they did not establish general convergence to the Euler equations, nor did they prove the consistency of arbitrary network formulations. A complete analysis would require explicit assumptions concerning point distributions, regularity, boundary conditions, refinement, and the convergence of the reconstructed physical fields.

Preliminary numerical investigations were conducted to assess the consistency of the pressure-gradient formulation and the relationship between virtual-volume evolution and continuum velocity divergence. Under selected joint-refinement conditions, both formulations exhibited convergence, whereas persistent discretization errors were observed under fixed kernel-to-spacing ratios. These results provide limited numerical evidence of consistency for the investigated one-dimensional configurations but do not establish the convergence or stability of a complete CSP solver (see Supplementary Material S2).

3.3. Shock Waves and Discontinuities

A major motivation of CSP was the possibility of representing sharp discontinuities without relying on conventional cell-based field reconstruction. However, shock waves are weak solutions governed by conservation laws and appropriate admissibility conditions. A computational method must reproduce physically correct propagation speeds and state changes, not merely maintain visually sharp interfaces.

For compressible flows, appropriate numerical behavior must be assessed against conservation jump conditions, entropy requirements, and reference solutions. No complete demonstration was obtained showing that CSP could reproduce the Rankine–Hugoniot conditions, select physically admissible shock solutions, or maintain stable behavior in the presence of strong discontinuities.

Consequently, the anticipated shock-capturing advantages of CSP remain unverified.

3.4. Numerical Stability and Computational Efficiency

The proposed structure was initially expected to offer possible benefits related to numerical stability, diffusion, and computational efficiency. However, these expectations were not established through a validated solver or systematic numerical comparison. An interaction-based computational system may encounter instability associated with point distribution, network singularities, force discretization, or time integration.

Similarly, computational cost depends on factors such as interaction searches, network construction, thermodynamic calculations, and numerical time stepping. A mesh-free or discrete-point representation does not necessarily require fewer computational operations than a mesh-based approach.

Therefore, no general conclusion regarding CSP's numerical stability, accuracy, or

computational efficiency relative to established CFD methods can be drawn from the available development.

4. Reassessment of Established CFD Methods

The original CSP investigation was partly motivated by perceived limitations of existing numerical approaches. Subsequent methodological assessment led to a more precise distinction between limitations that may occur in particular numerical schemes and limitations inherent to the governing equations themselves.

Conservative finite-volume methods, including Godunov-type shock-capturing schemes, provide established approaches for resolving compressible flows containing discontinuities [1]. Although a numerically captured shock may extend across several computational cells, this does not imply that its propagation speed, location, or associated conservation relationships must be inaccurate. Similarly, the existence of numerical instabilities in certain discretizations does not demonstrate that all mesh-based methods are unsuitable for shock-dominated flow analysis.

Particle-based methods also possess their own numerical strengths and limitations, which depend on their specific formulations and implementations [2, 3].

Accordingly, the comparative advantages initially anticipated for CSP cannot be justified solely by its avoidance of a conventional fixed-grid representation. Any such claim would require a complete implementation and controlled numerical comparisons under equivalent physical and computational conditions. This reassessment substantially reduced the justification for prioritizing the development of CSP over established methods for the immediate WDE research objectives.

5. Reasons for Research Suspension

In October 2026, further development of the CSP framework was suspended following approximately seven months of conceptual and methodological investigation.

The decision was based on several related considerations.

First, the framework remained mathematically incomplete. Although individual components and exploratory derivations had been developed, a general and internally consistent formulation suitable for numerical implementation had not been established.

Second, essential validation requirements remained unresolved. These included conservation under general conditions, numerical stability, continuum consistency, treatment of discontinuities, and physically admissible shock propagation.

Third, alternative formulations introduced additional conceptual complexity without providing a demonstrated resolution of the original difficulties.

Fourth, established finite-volume methods were found to provide a more immediately defensible computational foundation for investigating pressure-wave phenomena relevant to wave disk engines.

Finally, continued development of a new general-purpose CFD framework would require substantial additional theoretical and computational resources, potentially diverting attention from the underlying engineering questions.

The suspension should not be interpreted as proof that a CSP-based formulation is mathematically impossible or necessarily inferior to existing methods. Rather, it reflects the

absence of sufficient theoretical and computational evidence to justify continued prioritization of the framework under the present research objectives and available resources.

The CSP project is therefore classified as **Suspended — Conceptual and Unvalidated**.

6. Transition to Finite-Volume CFD for Hydrogen-Fueled Wave Disk Engines

The suspension of CSP does not eliminate the engineering problems that motivated its development.

Wave disk engines involve highly unsteady compressible flows, including pressure-wave propagation, shock interactions, expansion processes, and combustion-related pressure changes. These phenomena remain relevant to assessing the potential performance and limitations of hydrogen-fueled wave disk engines. Subsequent research will therefore focus on applying established conservative finite-volume methods to these physical problems.

The planned computational investigations will proceed through progressively extended models:

Stage I — One-Dimensional Hydrogen WDE Channel Analysis

Development and validation of a one-dimensional compressible flow solver, followed by the incorporation of hydrogen reaction kinetics and the analysis of combustion-driven pressure-wave behavior.

Stage II — Two-Dimensional Hydrogen WDE Channel Analysis

Extension to two spatial dimensions to investigate transverse flow structures, spatial nonuniformity, and the limitations of one-dimensional approximations.

Stage III — Two-Dimensional Hydrogen WDE Analysis with Port Effects

Incorporation of inlet and outlet port effects to examine interactions between wave propagation, reactive flow, and time-dependent flow boundaries. These stages represent planned research directions rather than completed investigations.

The selection of finite-volume methods is based on their established conservative formulation, availability of validated shock-capturing schemes, and suitability for systematic numerical verification. It does not constitute proof that CSP is incapable of producing accurate solutions.

The transition reflects a change in research priority: from developing a new general-purpose computational representation to investigating specific physical and engineering phenomena using established numerical tools.

7. Conclusion

The Computational State Point framework was investigated as a possible alternative representation for fluid dynamics based on discrete computational points, network relationships, and reconstructed thermodynamic states. The investigation produced several conceptual components and exploratory mathematical formulations, including virtual-volume

allocation and energy-based equations of motion.

However, the framework remained incomplete, and essential properties required for a reliable CFD method were not sufficiently established. In particular, no validated evidence was obtained demonstrating superior shock capturing, numerical stability, or computational efficiency compared with established methods.

Following a methodological reassessment, further CSP development was suspended in October 2026. The investigation will be preserved within the IGUER Research Archive as a record of conceptual development, unresolved scientific questions, and the rationale for redirecting subsequent research toward conservative finite-volume methods. The suspension does not preclude future reconsideration of the framework. Any renewed investigation would require a clearly specified mathematical model, rigorous conservation and consistency analysis, and numerical validation against established benchmark solutions.

Research Status: Suspended. No validated general-purpose CSP solver or demonstrated comparative performance advantage is claimed in this report.

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Appendix A. Previous CSP Research Report

The original 10-page CSP research report is preserved as supplementary material to this technical report. The document provides a historical record of the preliminary formulations and research objectives developed during the CSP project. Its contents reflect the state of the research at the time of preparation and should not be interpreted as validated findings.

Supplementary Material S1: Previous CSP Research Report (10 pages).

Appendix B. Preliminary Numerical Verification Records

This appendix provides the numerical verification report conducted during the development of CSP.

The report examines the consistency of the pressure-gradient formulation and the relationship between Virtual Volume evolution and continuum velocity divergence. The investigations identified convergence under selected refinement conditions, while also revealing limitations associated with discretization errors and kernel configurations.

These results are restricted to the investigated one-dimensional conditions and do not establish the general convergence, numerical stability, or shock-capturing performance of a complete CSP solver.

Supplementary Material S2: CSP Combined Numerical Verification Report (PDF).

Appendix C. Formal Verification Records Using Lean

This appendix contains four Lean source files developed during the mathematical investigation of CSP, with implementation assistance from OpenAI Codex.

The files cover inverse-distance and kernel-based network formulations, including mathematical statements concerning Virtual Volume, conservation-related relationships, pressure-force expressions, and conditional continuum-consistency results. They also contain counterexamples and analyses of limitations associated with the proposed formulations.

These records document exploratory mathematical formalization. The successful compilation of all source files has not been independently confirmed, and the encoded mathematical statements do not establish the physical validity or numerical reliability of a complete CSP solver.

Supplementary Material S3: Lean Formal Verification Records (ZIP containing four .lean files).

The supplementary materials are preserved as part of the IGUER Research Archive. Their inclusion does not imply independent peer review or validation of the complete CSP framework.

These materials constitute developmental research records and should not be interpreted as validated components of a completed numerical solver.

This technical report is published as part of the IGUER Research Archive. It documents preliminary and unvalidated research and has not undergone external peer review. The findings and formulations presented herein should not be interpreted as an established or validated computational method.